1 Estimation engine: the Kalman filter and RTS smoother

This section documents the recursive estimator at the core of the engine: a variable-dimension extended Kalman filter (EKF) with an optional prior-anchored iterated (IEKF) relinearisation pass, robust post-fit residual gating, and a backward Rauch–Tung–Striebel (RTS) smoother for post-processing modes. The state vector $\boldsymbol{x}$ is the uncombined PPP / PPP-RTK parameterisation of the state-vector-layout section of the Foundations document: receiver position (and optionally velocity), receiver clock and per-system time offsets, zenith wet troposphere and gradients, per-satellite slant ionosphere, receiver/satellite code and phase biases, and one float ambiguity per tracked carrier. Its dimension changes epoch to epoch as satellites rise and set and as ambiguities are reset; the layout and the dynamic key→index maps are owned exclusively by `filter/StateManager`, and the filter operates on that state without any fixed-index assumptions (`src/core/filter/KalmanFilter.cpp`).

Throughout we write $\boldsymbol{F}$ for the state-transition matrix, $\boldsymbol{Q}$ for the discrete process-noise covariance, $\boldsymbol{R}$ for the measurement-noise covariance, $\boldsymbol{S}$ for the innovation covariance, $\boldsymbol{K}$ for the Kalman gain, and $\boldsymbol{\nu}$ for the innovation (measurement minus predicted measurement). Predicted (a-priori) and updated (a-posteriori) quantities use the tilde/hat and $k|k-1$ / $k|k$ conditioning convention fixed in the notation-and-conventions section of the Foundations document.

1.1 Time update

State transition. The transition matrix $\boldsymbol{F}(\Delta t)$ for epoch interval $\Delta t = t_k - t_{k-1}$ is identity plus a small number of coupling blocks, assembled by `StateManager::getTransitionMatrix` (`src/core/filter/StateManager.cpp`). The kinematic-integrator off-diagonal couplings are

$$\boldsymbol{F}[\,\boldsymbol{r},\boldsymbol{v}\,] = \Delta t\,\boldsymbol{I}_3, \qquad \boldsymbol{F}[\,c\mathrm{d}t,\ c\dot{\mathrm{d}}t\,] = \Delta t, \qquad \boldsymbol{F}[\,c\delta t^{(g)},\ c\dot{\delta t}^{(g)}\,] = \Delta t, \tag{1}$$

present only when the corresponding rate state exists (`hasVelocity()`, clock-rate, per-system rate). A fourth coupling appears when the constant-velocity slant-iono model is armed (`estimation.enable_iono_rate`, default off): a per-satellite level/rate integrator

$$\boldsymbol{F}[\,I^i,\ \dot{I}^i\,] = \Delta t \qquad \text{for every non-frozen-pivot satellite,} \tag{2}$$

realising $I^i(k{+}1) = I^i(k) + \Delta t\,\dot{I}^i(k)$. Every remaining state, meaning ambiguities, zenith wet delay, gradients, code and satellite biases, any rate-free inter-system offset, and (absent `enable_iono_rate`) the per-satellite ionosphere, has $\boldsymbol{F} = \boldsymbol{I}$ and is propagated as a pure random walk with its mean held constant. Under the BSSD (single-difference) parameterisation the receiver-clock, phase-clock (`getPhaseClockIndex`), inter-system-offset, and frozen-pivot per-satellite ionosphere (`getFrozenPivotSatellites`) rows are unobservable (their $\boldsymbol{H}$ columns are zero); their position rows in Equation (1) are suppressed and, for each, the corresponding cross-covariances are zeroed in place after propagation while the diagonal variance is preserved (the `freeze_state` block in `src/core/filter/KalmanFilter.cpp`). The iono level/rate coupling of Equation (2), by contrast, is deliberately *not* gated off under BSSD: the slant-iono level remains observable under single-differencing (only the pivot column is zeroed), so every non-frozen-pivot satellite keeps its integrator; the frozen-pivot satellite alone is skipped, holding its rate at $(0,0)$ with its process noise suppressed so a live coupling cannot re-drive the pivot level off the SD datum.

The propagation itself is the standard EKF time update

$$\boxed{\begin{aligned}\tilde{\boldsymbol{x}}_{k|k-1} &= \boldsymbol{F}(\Delta t)\,\hat{\boldsymbol{x}}_{k-1|k-1}, &(3)\\ \tilde{\boldsymbol{P}}_{k|k-1} &= \boldsymbol{F}(\Delta t)\,\boldsymbol{P}_{k-1|k-1}\,\boldsymbol{F}(\Delta t)^{\mathrm{T}} + \boldsymbol{Q}(\Delta t), &(4)\end{aligned}}$$

implemented in `KalmanFilter::predict`. A non-positive Δt is treated as a no-op (duplicate or out-of-order epoch), short-circuiting before Equation (4).

Process-noise model. The discrete process noise $\boldsymbol{Q}(\Delta t)$ is built from continuous spectral densities by `StateManager::getProcessNoiseMatrix`. Two block structures appear. Pure random-walk states have a scalar discrete variance increment

$$\boldsymbol{Q}_{ii} = q_i\,\Delta t, \tag{5}$$

where q_i is the continuous variance rate (m^2/s) read from `filter.process_noise.*`. Representative entries are zenith wet troposphere (`troposphere_q`), the receiver code bias (`code_bias_q`), the satellite *code* bias (`sat_bias_q`, applied to `StateManager::sat_bias_indices_`), the satellite *phase* bias (`sat_phase_bias_q`, a separate density on the separate `sat_phase_bias_indices_` map), phase clock (`phase_clock_q`), and the float ambiguity (`ambiguity_q`). Both satellite-bias blocks, and the satellite-clock block beside them, suppress $\boldsymbol{Q}$ entirely for a frozen-pivot satellite, for the same reason as the freeze discussion above: the frozen state is held at (value, variance) $= (0, 0)$ as the datum, and any process noise would walk it off that datum. The ambiguity density defaults to a numerical safety floor (10^{-12}) rather than a physical value, and per-key overrides are supported for arc-specific reseeding. The per-satellite slant ionosphere uses a random walk whose zenith density is mapped to the slant by the squared obliquity factor,

$$\boldsymbol{Q}_{ii} = \frac{q_{\mathrm{iono}}}{\sin^2\theta^i}\,\Delta t, \qquad \sin^2\theta^i \text{ floored at } 10^{-2}\ (\theta \approx 5.7^\circ), \tag{6}$$

with q_{iono} = `ionosphere_q` and optional per-satellite overrides driven by the adaptive-iono scheme. Under the constant-velocity model of Equation (2) this random walk survives only as the fallback for satellites without a companion rate state: otherwise the level/rate pair takes the CWNA block of Equation (7) below with obliquity-scaled acceleration PSD $q = q_{\dot{\mathrm{iono}}}/\sin^2\theta^i$, where $q_{\dot{\mathrm{iono}}}$ = `process_noise.iono_rate_psd` (an acceleration PSD, $\mathrm{m}^2/\mathrm{s}^3$, default 10^{-10}). The adaptive-monitor override enters this branch as the *dimensionless* ratio $q_{\mathrm{ov}}/q_{\mathrm{iono}}$ multiplying the rate PSD, so the Sage–Husa storm response drives $\dot{I}^i$ rather than the level.

Coupled position/velocity, clock/clock-rate, inter-system offset/rate, and, under `enable_iono_rate`, slant-iono level/rate pairs use the continuous white-noise-acceleration (CWNA, integrated-random-walk) discretisation. For a coupled pair $(\xi, \dot{\xi})$ driven by acceleration PSD q,

$$\boldsymbol{Q}_{[\xi,\dot{\xi}]} = \begin{bmatrix} \frac{1}{3}\,q\,\Delta t^3 & \frac{1}{2}\,q\,\Delta t^2 \\ \frac{1}{2}\,q\,\Delta t^2 & q\,\Delta t \end{bmatrix}, \tag{7}$$

with q given by `acceleration_psd_q` (position/velocity, the 3×3 block-diagonal extension of Equation (7) over the ECEF axes), `clock_rate_q`, `system_time_rate_offset_q`, and the obliquity-scaled `iono_rate_psd`$/\sin^2\theta^i$ respectively. When the velocity state is present the rate-free position density `position_q` is unused. Baseline (double-difference) mode reduces $\boldsymbol{Q}$ to the position block, a per-ambiguity random-walk term, and an optional differential-ZWD entry $\boldsymbol{Q}_{ii} = \sigma^2\Delta t$ governed by `baseline.differential_tropo_process_noise_m_per_sqrt_s`.

Scope: the recursive estimator only. These densities govern the *recursive* estimator. The network producer runs no `KalmanFilter` at all: `getTransitionMatrix` and `getProcessNoiseMatrix` are reached only from `KalmanFilter::predict` / `predictImu` / `predictAided` and from the smoother's `Application::capturePriorSnapshot`. The producer instead solves per-epoch normal equations for corrections to each station's linearisation point, and imposes its cross-epoch stiffness with explicit information-form carry priors of effective variance $\mathrm{var}_{\mathrm{eff}} = \mathrm{var} + q\,\Delta t$ rather than through $\boldsymbol{Q}(\Delta t)$ (the network-mode document). The satellite-side densities `sat_clock_q`, `sat_phase_bias_q` and `sat_code_bias_q` are read there as exactly those carry rates, and are additionally authorable through the always-live network-scoped `network.process_noise` class table, whose per-class fallbacks equal the legacy `filter.process_noise.*` defaults so an unmodified config resolves identically either way.

Aided time update. When inertial aiding is active, `KalmanFilter::predictImu` performs a strapdown mechanisation of the nominal position, velocity, and attitude and splices a 15×15 INS error-state transition $\boldsymbol{\Phi}$ and noise $\boldsymbol{Q}_{\mathrm{ins}}$, with error order $[\delta\boldsymbol{p},\, \delta\boldsymbol{v},\, \delta\boldsymbol{\psi},\, \delta\boldsymbol{b}_a,\, \delta\boldsymbol{b}_g]$, into the GNSS-dimensioned $\boldsymbol{F}$ and $\boldsymbol{Q}$ over the integrated interval, after which Equation (4) runs on the full state with explicit symmetrisation $\boldsymbol{P} \leftarrow \frac{1}{2}(\boldsymbol{P} + \boldsymbol{P}^{\mathrm{T}})$. The generalised `predictAided` path applies the same splice for arbitrary aiding sources and delegates byte-identically to the GNSS-only Equation (4) when no aiding payload is present.

1.2 Measurement update

The measurement update (`KalmanFilter::update`) is a *batch* update: all observation rows of the epoch are processed simultaneously, not sequentially/scalar. A sparse-$\boldsymbol{H}$ optimisation accelerates the four design-matrix products ($\boldsymbol{HP}$, $\boldsymbol{HPH}^{\mathrm{T}}$, $\boldsymbol{KH}$, and the $\boldsymbol{H}$-touching parts of the gain), and is numerically identical to the dense computation to within a 10^{-12} structural threshold (`src/core/filter/KalmanFilter.cpp`).

Given the design matrix $\boldsymbol{H}$ from the linearisation of the code observation equation and phase observation equation of the Foundations document (position rows $-(\boldsymbol{e}^{i}_{\hat{r}})^{\mathrm{T}}$, the linearised geometric range there), the innovation $\boldsymbol{\nu}$ (`measurement.innovations`), and the measurement-noise covariance $\boldsymbol{R}$ (a full matrix when supplied, otherwise the diagonal $\boldsymbol{R} = \mathrm{diag}(\sigma^2_{r,i,s})$ of the stochastic noise model there), the innovation covariance and gain are

$$\boldsymbol{S} = \boldsymbol{H}\,\tilde{\boldsymbol{P}}_{k|k-1}\,\boldsymbol{H}^{\mathrm{T}} + \boldsymbol{R}, \tag{8}$$

$$\boldsymbol{K} = \tilde{\boldsymbol{P}}_{k|k-1}\,\boldsymbol{H}^{\mathrm{T}}\,\boldsymbol{S}^{-1}. \tag{9}$$

The gain is never formed by explicitly inverting $\boldsymbol{S}$. Exploiting the symmetry of $\boldsymbol{S}$, the engine factorises it once with an $\mathrm{LDL}^{\mathrm{T}}$ decomposition and recovers $\boldsymbol{K} = \left(\boldsymbol{S}^{-1}\boldsymbol{H}\tilde{\boldsymbol{P}}\right)^{\mathrm{T}}$ from a single solve against $\boldsymbol{H}\tilde{\boldsymbol{P}}$ (`ldlt.solve(HP).transpose()`); on factorisation failure it falls back to a complete-orthogonal (pseudo-inverse) solve. The state update is

$$\delta\boldsymbol{x} = \boldsymbol{K}\,\boldsymbol{\nu}, \qquad \hat{\boldsymbol{x}}_{k|k} = \tilde{\boldsymbol{x}}_{k|k-1} + \delta\boldsymbol{x}. \tag{10}$$

Joseph-form covariance update. The covariance is updated in the numerically stabilised *Joseph* form, never the short form $(\boldsymbol{I} - \boldsymbol{KH})\tilde{\boldsymbol{P}}$:

$$\boxed{\boldsymbol{P}_{k|k} = (\boldsymbol{I} - \boldsymbol{KH})\,\tilde{\boldsymbol{P}}_{k|k-1}\,(\boldsymbol{I} - \boldsymbol{KH})^{\mathrm{T}} + \boldsymbol{K}\,\boldsymbol{R}\,\boldsymbol{K}^{\mathrm{T}}.} \tag{11}$$

The Joseph form preserves symmetry and positive semi-definiteness for any gain, which matters because the engine deliberately departs from the optimal gain in its robust paths (Section 1.4); the result is symmetrised once more, $\boldsymbol{P} \leftarrow \frac{1}{2}(\boldsymbol{P} + \boldsymbol{P}^{\mathrm{T}})$, to clear round-off. The factor $(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})$ is assembled from the sparse $\boldsymbol{K}\boldsymbol{H}$ product with a unit shift on the diagonal. The same Joseph kernel, with the explicit gain $\boldsymbol{K} = \boldsymbol{P}\boldsymbol{H}^{\mathrm{T}}\boldsymbol{S}^{-1}$, is exposed as the pure static routine `josephPseudoUpdate` and reused for scalar pseudo-observation constraints (e.g. the GIM ionosphere prior via `applyPseudoObservation`) and for the IEKF pass of Section 1.3.

Post-fit residuals. After the update the post-fit residuals and their (diagonal) covariance are

$$\boldsymbol{r} = \boldsymbol{\nu} - \boldsymbol{H}\,\delta\boldsymbol{x}, \qquad S_{r,ii} = R_{ii} - \left(\boldsymbol{H}\,\boldsymbol{P}_{k|k}\,\boldsymbol{H}^{\mathrm{T}}\right)_{ii}, \tag{12}$$

of which, by default, only the diagonal of S_r is materialised. When the off-diagonal-R kernel is enabled (`enable_full_r_kernel`, default off) *and* a dense $\boldsymbol{R}$ was supplied (BSSD / baseline-DD epochs whose $\boldsymbol{R}$ carries SD/DD cross-covariance), the full symmetrised $S_r = \boldsymbol{R} - \boldsymbol{H}\,\boldsymbol{P}_{k|k}\,\boldsymbol{H}^{\mathrm{T}}$ is instead formed and stored, so the redundancy and w-test QC of Section 1.4 honour correlated observations via S_r^{-1}; the diagonal, and hence Equation (13), is identical either way. The standardised residual that drives gating (Section 1.4) is

$$s_i = \frac{r_i}{\sqrt{\max\{\, S_{r,ii},\, R_{ii},\, 10^{-12} \,\}}}. \tag{13}$$

The denominator floor at the *working* R_{ii} (the inflated noise actually used in $\boldsymbol{R}$, not the formal noise) is a deliberate robustness compromise: in ambiguity-absorption cases (double-differenced phase with an ambiguity gain near unity) the rigorous denominator $S_{r,ii}$ collapses toward $R\,\boldsymbol{S}^{-1}R$, which would register near-zero residuals as many-σ outliers. The strictly correct standardisation uses $S_{r,ii}$ alone; the floor trades that rigour for stability and is documented as such in source.

1.3 Iterated relinearisation (IEKF)

For long initial position offsets a single linearisation of the geometric-range relation of the Foundations document per epoch can be insufficient. When `processing.relinearize_geometry` is enabled (and the run is not tightly-coupled IMU), the engine performs a prior-anchored iterated update, a Gauss–Newton / iterated EKF step, after the primary update.

Anchor. The iteration is anchored to the *stashed pre-update prior* $(\boldsymbol{x}^-, \boldsymbol{P}^-)$ that the converged primary Equations (4) and (11) read, captured only when relinearisation is armed (`getLastUpdatePriorX` / `getLastUpdatePriorP`). This is the prior *after* any in-epoch ambiguity augmentation and outlier reset, so it is dimension-matched to the posterior; it is not a top-of-loop snapshot.

Gauss–Newton step. Let $\boldsymbol{x}_i$ be the current linearisation iterate (the committed state at which `measurement` was linearised, with innovations $\boldsymbol{z} - h(\boldsymbol{x}_i)$ and $\boldsymbol{H} = \partial h/\partial \boldsymbol{x}\big|_{\boldsymbol{x}_i}$). The prior-anchored iterate (`KalmanFilter::updateFromPrior`, via `josephPseudoUpdate`) is

$$\boxed{\begin{aligned} \boldsymbol{\nu} &= \left(\boldsymbol{z} - h(\boldsymbol{x}_i)\right) - \boldsymbol{H}(\boldsymbol{x}^- - \boldsymbol{x}_i), && (14)\\ \boldsymbol{K} &= \boldsymbol{P}^-\boldsymbol{H}^{\mathrm{T}}\left(\boldsymbol{H}\boldsymbol{P}^-\boldsymbol{H}^{\mathrm{T}} + \boldsymbol{R}\right)^{-1}, \qquad \boldsymbol{x}^+ = \boldsymbol{x}^- + \boldsymbol{K}\,\boldsymbol{\nu}, && (15)\\ \boldsymbol{P}^+ &= (\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\,\boldsymbol{P}^-\,(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})^{\mathrm{T}} + \boldsymbol{K}\,\boldsymbol{R}\,\boldsymbol{K}^{\mathrm{T}}. && (16) \end{aligned}}$$

The offset term $-\boldsymbol{H}(\boldsymbol{x}^- - \boldsymbol{x}_i)$ in Equation (14) re-expresses the innovation about the fixed prior and is what makes this a proper MAP / Gauss–Newton iteration: because $\boldsymbol{K}$ and $(\boldsymbol{I} - \boldsymbol{K}\boldsymbol{H})\boldsymbol{P}^-$ are always derived from the same fixed prior $(\boldsymbol{x}^-, \boldsymbol{P}^-)$, the covariance does *not* contract a second time and no information is double-counted.

Firing condition. The driver `Application::runGeometryRelinearization` (`src/core/app/Application.cpp`) fires the iteration when all of the following hold:

1. `processing.relinearize_geometry` is on and a prior was stashed at the primary update;
2. the position change since the primary update exceeds a data-dependent threshold, `relinearize_dpos_threshold_phase_m` (default 1 m) when phase is present, or `relinearize_dpos_threshold_code_only_m` (default 100 m) for code-only epochs;
3. the per-run cap `relinearize_max_count` (-1 = unlimited) is not exhausted.

The inner loop recomputes geometry and re-applies Equations (14) to (16) against the *fixed* converged exclusion set, for at most five iterations, declaring convergence when the position step falls below 10^{-4} m. Because GNSS geometry is nearly linear at the metre level, convergence is typically reached in a single pass and the threshold of item 2 means the iteration rarely fires at all. The tightly-coupled IMU configuration instead takes a single legacy *naive* second pass (re-applying the measurement against the posterior), since error-state attitude/bias feedback conflicts with a clean prior re-derivation.

1.4 Innovation and residual gating

The engine performs *no* χ^2 innovation-covariance test. Outlier rejection (`KalmanFilter::validatePostFitResiduals`) operates on the standardised *post-fit* residuals s_i of Equation (13) against a fixed scalar threshold, supplemented by robust median-based spatial detectors for pivot slips. The two top-level thresholds are hardcoded `GnssConfig` defaults (`residual_threshold_` $= 6.0$ and `bad_meas_percent_` $= 0.5$ in `src/core/config/GnssConfig.cpp`) that are deliberately not exposed to JSON: the per-row rejection threshold is $s_{\max} = 6.0$ (read at runtime via `getResidualThreshold`) and the complete-reset trigger fraction is 0.5 (`getBadMeasPercent`); there are no `residual_threshold_` or `bad_meas_percent_` config keys to search for. The dispatch row and criterion can nonetheless be *replaced* wholesale by the Baarda data-snooping gate below, itself JSON-exposed.

Per-row rejection. If $\max_i |s_i| > 6.0$, the worst row is dispatched by type: a phase outlier triggers an ambiguity reset (with BSSD pivot handling and a sub-cycle λ-fraction floor), while a code outlier is excluded from the epoch. Correlated same-satellite phase rows on other bands are reset alongside it.

Baarda data-snooping gate (optional override). When `estimation.enable_data_snooping` is set, the fixed-σ gate triple $(\text{gate stat}, \text{gate row}, \text{gate crit}) = (\max_i |s_i|, \arg\max, 6.0)$ is *replaced* in `validatePostFitResiduals` by the Baarda w-test: over *checkable* rows, those whose redundancy $r_i = S_{r,ii}/R_{ii}$ meets a floor ($r_i \geq 0.1$; low-redundancy rows are skipped as uncheckable, since a blunder there is absorbed into the estimate rather than the residual), the worst $|w_i|$ is selected and the gate fires on $|w_i| >$ `estimation.data_snooping_critical` (default 3.29). Both the criterion *and* the dispatched row can thus differ from the σ gate; the type-based dispatch of the selected row is otherwise unchanged. The w-test statistic and redundancy are produced by the a-posteriori

QC diagnostics layer (`computeResidualQc_`, gated by `estimation.enable_variance_factor`): the variance factor $\hat{\sigma}_0^2 = \big(\sum_i v_i^2/R_{ii}\big)/\mathrm{dof}$, the per-row redundancy r_i, the w-test, and the minimal detectable bias (MDB). This layer honours a correlated R through S_r^{-1} when the full-R kernel is active (see Equation (12)).

Complete-reset backstop. If the count of rejected rows exceeds $0.5\times$ (original pre-peel observation count), the filter covariance is re-inflated (`resetFilterCovariance`), a hard reset that, in the smoother, marks a segment boundary (Section 1.5).

Robust pivot-slip detectors (BSSD). Before the per-row gate, single-difference phase rows are grouped by (pivot, signal) and screened for a coherent pivot slip using median / median-absolute-deviation (MAD) statistics: a forced pivot handover fires when a group is large enough, a majority of its rows exceed threshold, the group median exceeds a metre-scale floor, and the MAD-to-median ratio is small (a tight cluster indicating a common-mode shift rather than scattered noise). Cross-band pooled and code-side variants apply analogous tests with their own floors, and non-pivot multi-band slips are confirmed by integer-cycle consistency ($|r|/\lambda_{\mathrm{band}}$ rounding to the same nonzero integer across bands).

Down-weighting (not gating). Two paths inflate $\boldsymbol{R}$ rather than reject: a persistent EWMA-bias term adds the squared mean bias to R_{ii} once an arc has warmed up, and a Huber post-fit scheme inflates R for sub-cycle rows whose $|s_i|$ exceeds the Huber knee (innovation one-shot or post-fit IRLS variants, toggled by `enable_huber_r_inflation` and `enable_huber_innovation`). These modify the working $\boldsymbol{R}$ that feeds Equations (8) and (13), and are the reason the standardised residual floors at the working noise rather than the formal noise.

1.5 Backward RTS smoother

In post-processing modes, enabling `processing.run_smoother` runs a fixed-interval Rauch–Tung–Striebel recursion [1] over the stored forward trajectory (`include/filter/Smoother.hpp`, `src/core/filter/Smoother.cpp`). The flag is force-disabled under `AppMode::RealTime`, since smoothing is non-causal and inherently a post-processing operation.

Forward-pass capture. Rather than reconstruct the a-priori from $\boldsymbol{F}$ and $\boldsymbol{Q}$ on the backward pass, the forward pass caches, per epoch, both the one-step prediction $\tilde{\boldsymbol{x}}_{k|k-1}, \tilde{\boldsymbol{P}}_{k|k-1}$ and the posterior $\hat{\boldsymbol{x}}_{k|k}, \boldsymbol{P}_{k|k}$, together with $\boldsymbol{F}_k$ and the full set of core and dynamic key→index maps (`capturePriorSnapshot` / `capturePosteriorSnapshot` in `src/core/app/Application.cpp`). A capture-ordering invariant makes the stored prior exactly $\boldsymbol{F}_k\hat{\boldsymbol{x}}_{k-1|k-1}$: the a-priori is recorded immediately after `predict()` and before any cycle-slip reseeding or measurement update, and the a-posteriori is recorded *before* integer ambiguity resolution, so the smoother consumes *float* ambiguities only.

Recursion. Initialised at the terminal posterior $\hat{\boldsymbol{x}}_{N|N} = \hat{\boldsymbol{x}}_{N|N}$, $\boldsymbol{P}_{N|N} = \boldsymbol{P}_{N|N}$, the backward recursion is

$$\boxed{\begin{aligned} \boldsymbol{G}_k &= \boldsymbol{P}_{k|k}\,\boldsymbol{F}_{k+1}^{\mathrm{T}}\,\tilde{\boldsymbol{P}}_{k+1|k}^{-1}, && (17)\\ \hat{\boldsymbol{x}}_{k|N} &= \hat{\boldsymbol{x}}_{k|k} + \boldsymbol{G}_k\big(\hat{\boldsymbol{x}}_{k+1|N} - \tilde{\boldsymbol{x}}_{k+1|k}\big), && (18)\\ \boldsymbol{P}_{k|N} &= \boldsymbol{P}_{k|k} + \boldsymbol{G}_k\big(\boldsymbol{P}_{k+1|N} - \tilde{\boldsymbol{P}}_{k+1|k}\big)\boldsymbol{G}_k^{\mathrm{T}}. && (19) \end{aligned}}$$

The smoother gain $\boldsymbol{G}_k$ is realised as an $\mathrm{LDL}^{\mathrm{T}}$ solve: the predicted covariance is symmetrised, then $\boldsymbol{G}_k = \big(\tilde{\boldsymbol{P}}_{k+1|k}^{-1}(\boldsymbol{P}_{k|k}\boldsymbol{F}^{\mathrm{T}})^{\mathrm{T}}\big)^{\mathrm{T}}$ via `ldlt.solve(...).transpose()`, never an explicit inverse, and the smoothed covariance Equation (19) is symmetrised after the update. If the factorisation fails, the step is abandoned and epoch k keeps its forward posterior.

Variable-dimension reconciliation. Because $\dim \boldsymbol{x}_k \neq \dim \boldsymbol{x}_{k+1}$ in general, Equations (17) to (19) are restricted at each step to the *common-key subspace*, the states present at both k and $k{+}1$, found by intersecting the dynamic key→index maps (`collectCommonIndices`, less any keys in the handover exclusion set of the segment-boundary paragraph below). Let $\boldsymbol{\Pi}_k$ and $\boldsymbol{\Pi}_{k+1}$ select the common rows in each epoch's state space; the implemented state update is then

$$\hat{\boldsymbol{x}}_{k|N} = \hat{\boldsymbol{x}}_{k|k} + \boldsymbol{\Pi}_k^{\mathrm{T}}\,\boldsymbol{G}_{k,C}\big(\boldsymbol{\Pi}_{k+1}\hat{\boldsymbol{x}}_{k+1|N} - \boldsymbol{\Pi}_k\tilde{\boldsymbol{x}}_{k+1|k}\big), \tag{20}$$

with the analogous projection on Equation (19). States outside the subspace retain their forward a-posteriori value. A subtle indexing invariant: $\tilde{\boldsymbol{x}}_{k+1|k}$, $\tilde{\boldsymbol{P}}_{k+1|k}$ and $\boldsymbol{F}_{k+1}$ are sized for epoch k's layout (the predict step does not grow the state), so they are addressed with epoch k's common indices, whereas $\hat{\boldsymbol{x}}_{k+1|N}$, $\boldsymbol{P}_{k+1|N}$ live in epoch $k{+}1$'s layout.

Segment boundaries at resets and frame changes. Four mechanisms of differing severity terminate or trim backward information flow at ambiguity resets, teardowns, and pivot handovers:

- A *complete* filter reset at $k{+}1$ (the covariance re-inflation of Section 1.4, recorded as `is_discontinuity`) is a hard segment boundary: it severs the dynamic link $\tilde{\boldsymbol{x}}_{k+1|k} = \boldsymbol{F}\hat{\boldsymbol{x}}_{k|k}$, so no $\boldsymbol{G}_k$ is meaningful and backward flow is terminated; epoch k keeps its forward posterior.
- A *single-satellite* ambiguity reset is handled softly: that satellite's ambiguities are excluded from the common subspace for the $k \to k{+}1$ step, because the re-seeded large-variance prior at $k{+}1$ carries no information backward and smoothing it would only inject noise. The exclusion matches by satellite prefix and is intentionally slightly over-broad, always erring toward the safe forward fallback.
- A *dimension change* $\dim \tilde{\boldsymbol{x}}_{k+1|k} \neq \dim \hat{\boldsymbol{x}}_{k|k}$ is a hard boundary: predict never grows the state, so any size mismatch signals a pre-predict long-gap teardown (`Application::handleGap` prunes ambiguity / iono / sat-bias states before `predict()`), across which epoch k's indices would address the wrong states in the $k{+}1$ a-priori, so backward flow stops and epoch k keeps its forward posterior.
- A *BSSD pivot handover* between k and $k{+}1$ re-references one or more constellations in place; it is reconciled or segmented per constellation as described next.

BSSD pivot-handover reconciliation. When the pivot map changes between epochs ($\texttt{pivot_satellites}_k \neq \texttt{pivot_satellites}_{k+1}$) the smoothed posterior at $k{+}1$ sits in the *new* pivot frame while the cached a-priori $\tilde{\boldsymbol{x}}_{k+1|k}$ (captured pre-handover, in epoch k's layout) is still in the *old* frame, so the two are not directly comparable in Equation (18). `Smoother::prepareHandoverFrame` restores comparability per constellation on *copies* of the $k{+}1$ smoothed estimate (leaving the reported $k{+}1$ result in its own frame). A handover recorded as *clean* by `PivotManager` (old/new pivots matching `clean_handovers`) is inverted back to the old pivot p_0 signal-group by signal-group through the public static `Smoother::inverseReReferenceGroup`, the exact inverse of the forward re-reference. For a group with old-pivot datum row d,

$$x'(i) = x(i) - x(d), \quad x'(d) = 0, \qquad P'(i,\cdot) = P(i,\cdot) - P(d,\cdot), \quad P'(\cdot,j) = P(\cdot,j) - P(\cdot,d), \tag{21}$$

applied over *full* rows and columns (the forward re-reference coupled the group to every other state, so the inverse must too) with the datum row/column zeroed last. A signal group is invertible only if the old pivot still carries a live state on it *and* the forward leg re-referenced the signal to the new pivot; orphaned groups (the new pivot lacked the signal), disturbed or forced handovers, and every handover constellation's ionosphere and satellite-bias states are instead *segmented*, added to the `exclude_keys` set that `collectCommonIndices` drops from the common subspace, so those states fall back to the forward posterior.

Static-position degeneracy. In static PPP the receiver position is a (near-)constant state, so the position-block smoother gain propagates essentially the same correction to every epoch and the smoothed position is degenerate, constant across epochs, *by construction*. This is expected, not a defect: the smoother's value accrues on the non-position states (troposphere, ionosphere, float ambiguities) and on kinematic runs where position carries genuine epoch-to-epoch dynamics.

References

[1] H. E. Rauch, F. Tung, and C. T. Striebel. "Maximum likelihood estimates of linear dynamic systems". In: *AIAA Journal* 3.8 (1965), pp. 1445–1450. DOI: 10.2514/3.3166.